Integration of multi-omics and immunogenicity data enhances the classification of cancer phenotypes

Meihong Jin[1], Kwoneel Kim[2]

[1] Department of Biomedical and Pharmaceutical Sciences, Kyung Hee University, Seoul, Korea

[2] Department of Biology, Kyung Hee University, Seoul, Korea

Abstract

There is an urgent need for integrated data model addressing both cancer multi-omics and tumor immune microenvironment features. Here, we constructed an integrated data model including multi-omics and immunogenicity from large cancer cohort databases that were TCGA, PCAWG, and TCIA. We curated the data by applying uniformed normalization strategy and validated that the normalization was effective in clustering normal and cancer groups. Furthermore, we tried to predict the cancer phenotypes of metastasis and MSI status using the integrated data model. The prediction model constructed based on multi-omics data showed better classification performance than the model of single-omics data. Also, the model based on both multi-omics and immunogenicity data had the best prediction performance in consonance with different cancer types. The ranks of features acting importantly in the prediction were different across cancer types, which indicate that explanatory power of each feature was dependent on the context of relevant cancer type. These results imply that cancer phenotype could be more predictable when we consider multi-omics and tumor immune microenvironment together.

Methods

Data acquisition

We downloaded the genomics data, transcriptome, proteome data, methylome data, and clinical data of normal and tumor samples of TCGA and PCAWG samples from the UCSC Xena database (https://xenabrowser.net) Additional immunogenicity data such as immunophenoscore was downloaded from TCIA (https://tcia.at/patients). Other clinical data were downloaded from previous publications [1, 2]. Finally, 12,356 and 3,312 samples were obtained in TCGA and PCAWG datasets, respectively.

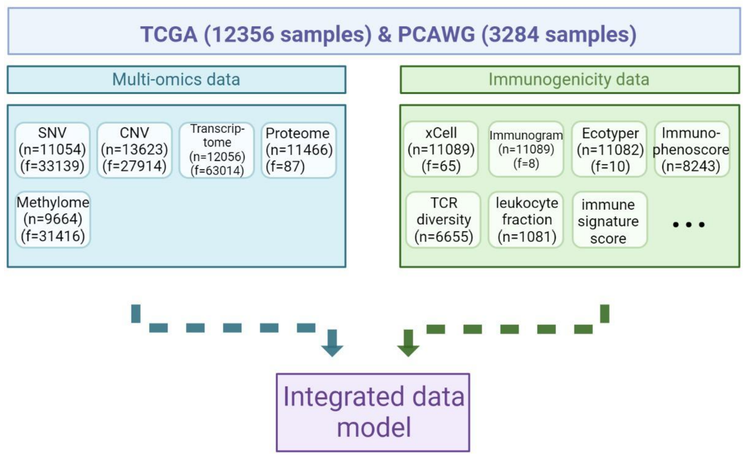


Figure 1. Construction of the database

Immunogenicity data processing

Transcriptome data of tumor samples from TCGA and PCAWG cohorts were involved in the processing of immunogenicity data. Identification of cell states and cellular communities was performed with EcoTyper (https://ecotyper.stanford.edu/carcinoma/) in an unsupervised manner [3]. xCell and immunogram scores were calculated and samples were classified into two phenotypes, HTE and CTE tumors [4, 5, 6].

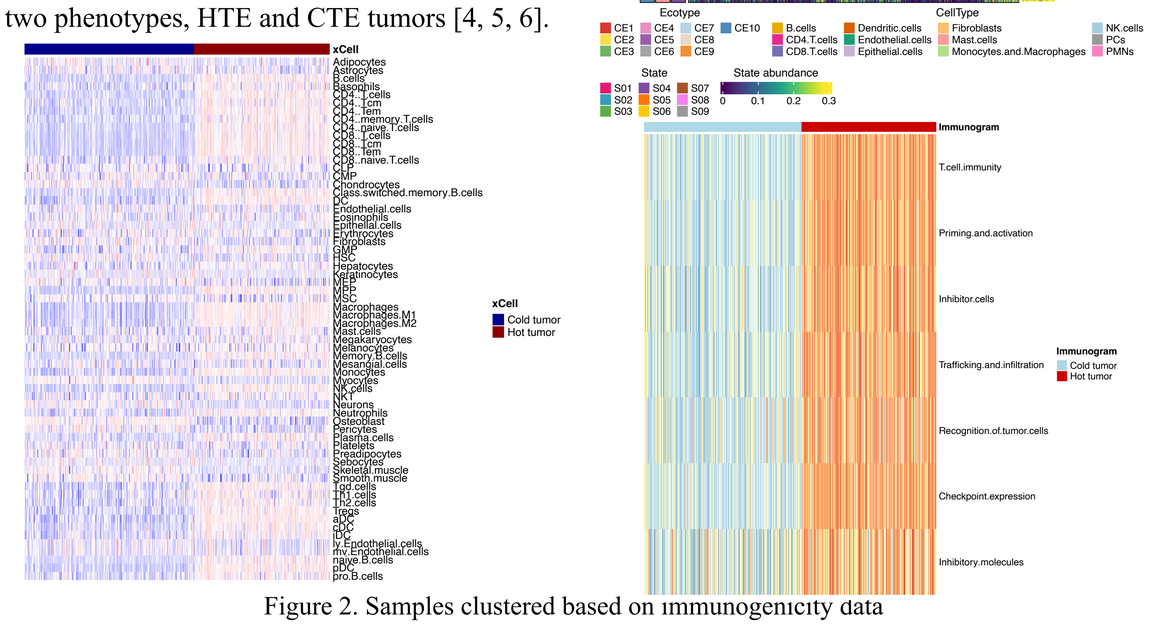


Figure 2. Samples clustered based on immunogenicity data

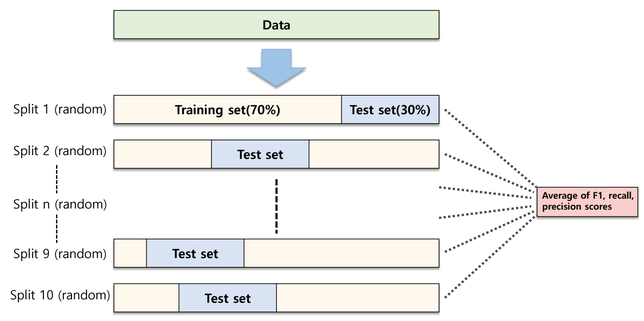
Normal-Tumor clustering after normalization

Gene expression and methylation data of cancer and normal samples across these 5 cancer types were processed into single-omics and multi-omics manner. We then performed gene-wise quantile normalization and clustering analysis with K-means and hierarchical algorithm, which was implemented in the database. For the comparison, the same analysis was performed on raw data without normalization. One proportion Z-tests were carried out on clustering results using the prop.test function in R.

Performance evaluation of machine learning algorithms

The targeted clinical feature was metastasis status and MSI status. We ran four types of classification models, support vector machine (SVM), multi-layer perceptron (MLP) classifier, random forest algorithm, and k-nearest neighbors algorithm (k-NN) with processed final files. F1, precision, and recall scores were calculated and returned along with other results. Then, in an attempt to further clarify the most important features in each merged data, we performed a feature importance analysis with a random forest model.

Figure 3. Proposed architecture of classification models.



Results

The normalization of single- and multi-omics data enhances clustering of normal and cancer groups

BRCA, HNSC, and UCEC showed better clustering results of normal and cancer groups when the normalization was performed although some features did not have statistical significance. (Fig. 4A, B). However, KIRC and LUAD had better clustering results with normalization only in transcriptome data and no significant clustering results in methylome data. We hypothesized that this poor clustering performance based on single-omics data in some cancer types because the specific single-omics had insufficient explanatory power to divide the normal and cancer samples. Therefore, we performed clustering analysis for the methylome data of KIRC and LUAD with constructing multi-omics data by integration of transcriptome and a set of multi-omics data showed better clustering results of normal and cancer groups than the clusters based on single-omics data (Fig. 4C).

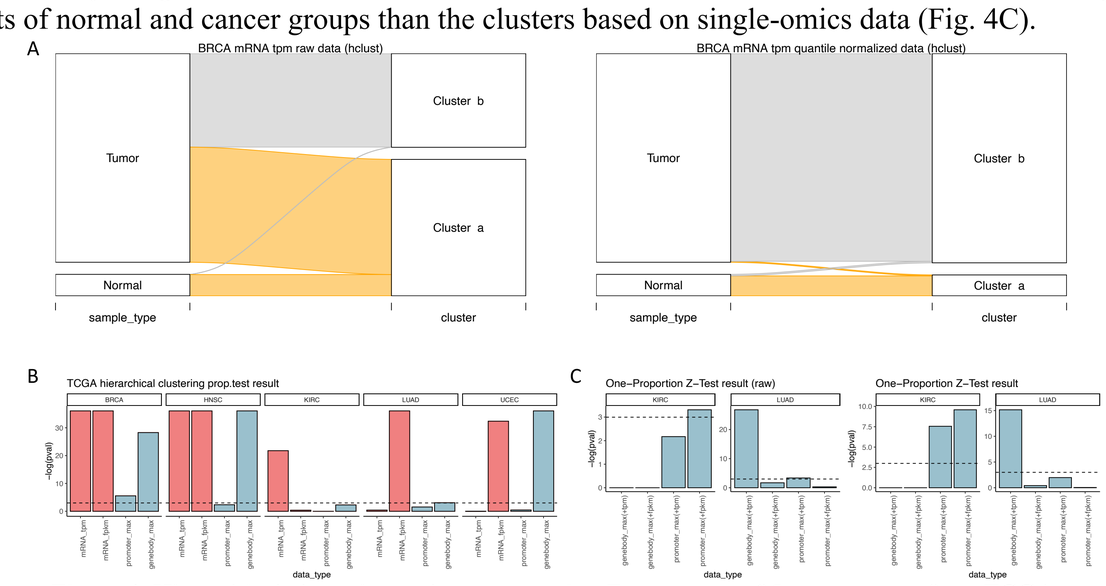


Figure 4. Hierarchical clustering and one proportion Z-test resuts of 5 cancer types from TCGA.

Integration of cancer immunogenicity data to multi-omics data enhances the performance of metastasis prediction

The prediction performance of the model constructed using multi-omics data was better than the model based on single-omics data (orange box vs light blue box in Fig. 5A) regardless of cancer types, performance measures, and machine learning algorithms. Furthermore, the integration of immunogenicity data to the models based on single-omics and multi-omics data enhanced the prediction performance in consistent with overall predictions (dark-colored box vs light-colored box in Fig. 5A). The model constructed using both of multi-omics and immunogenicity data showed the best and the most robust prediction performance. Top 30 features were obtained from the feature importance analysis and various types of omics and immunogenicity data were ranked in the five cancer types (Fig. 5B). Several high rank features were reported to be associated with metastasis mechanism and BRCA and LUAD were explained more by immunogenicity features than other cancer types consistently using different numbers of high rank features (Fig. 5C).

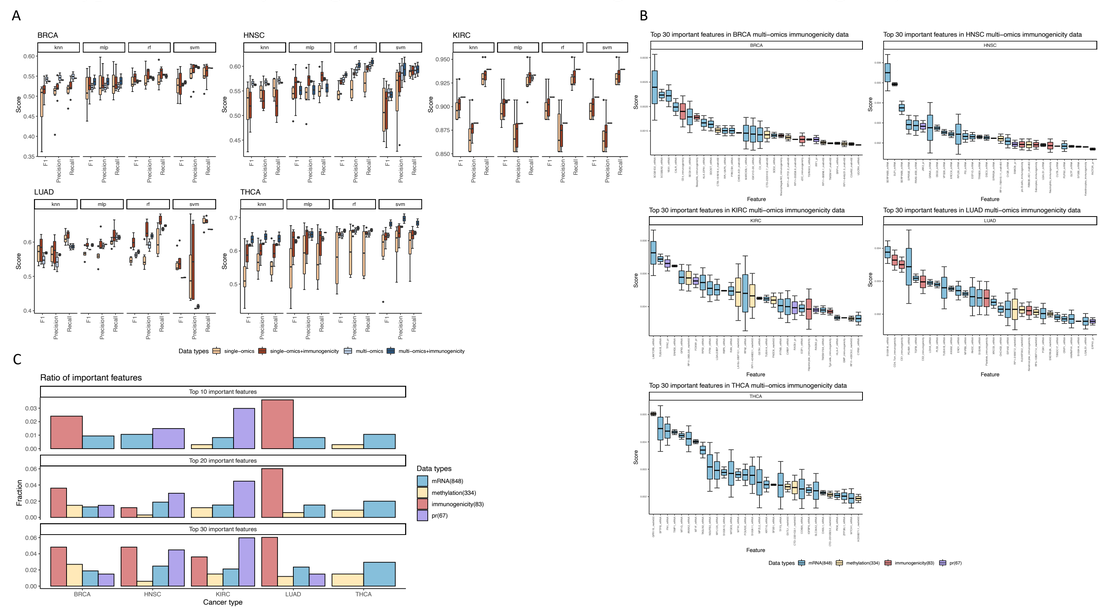


Figure 5. Performance evaluation of 4 classification algorithms of 5 cancer types from TCGA.

Conclusion

We constructed an integrated data model including multi-omics and immunogenicity data from the largest cancer cohorts of TCGA, PCAWG, and TCIA. We firstly confirmed that the continuous degree of expression or methylation in multi-omics data was needed to be normalization in the clustering result of normal and cancer groups and the importance of multi-omics approach for the clustering study in cancer. We analyzed the validity of integrating multi-omics and immunogenicity data by testing the performance of prediction model, and the prediction model showed best performance in prediction of metastasis and MSI status when it was composed by both multi-omics and immunogenicity data. This result implies that cancer phenotype and molecular characteristic could be more predictable when we consider multi-omics regulation and tumor immune microenvironment together.

References

[1] Kim, K. *et al.* Predicting clinical benefit of immunotherapy by antigenic or functional mutations affecting tumour immunogenicity. *Nat Commun* 11, (2020).

[2] Thorsson, V. *et al.* The Immune Landscape of Cancer. *Immunity* 48, 812-830.e14 (2018).

[3] Luca, B. A. *et al.* Atlas of clinically distinct cell states and ecosystems across human solid tumors. *Cell* 5482–5496 (2021) doi:10.1016/j.cell.2021.09.014.

[4] Aran, D., Hu, Z. & Butte, A. J. xCell: Digitally portraying the tissue cellular heterogeneity landscape. *Genome Biol* 18, 1–14 (2017).

[5] Satpathy, S. *et al.* A proteogenomic portrait of lung squamous cell carcinoma. *Cell* 184, 4348-4371.e40 (2021).

[6] Karasaki, T. *et al.* An Immunogram for the Cancer-Immunity Cycle: Towards Personalized Immunotherapy of Lung Cancer. *Journal of Thoracic Oncology* 12, 791–803 (2017).